SUDEEPA N S

Software Development Engineer in Test (SDET)

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PROFESSIONAL SUMMARY

SDET with 1.9 years architecting enterprise automation frameworks in Java, Playwright, and Selenium. Expertise in designing intelligent client-side frameworks, query-based test data fetching, and Agentic AI integration with MCP. Proven ability to optimize performance through token reduction strategies and CLI-based deterministic operations. Track record of improving test reliability from 60% to 97% and scaling frameworks to 5,000+ test scenarios with zero data contamination across parallel execution.

EXPERIENCE

Software Development Engineer in Test (SDET) | Azentio Software | Bengaluru, India | Sep 2024 - Present

Framework Architecture & Design:

- Architected client-side framework with intelligent screen-scanning engine that dynamically identifies UI labels and generates label-ID mappings, reducing manual locator maintenance by 70%
- Designed query-based test data fetching mechanism eliminating hardcoded test data, enabling dynamic scenario execution across multiple environments
- Implemented comprehensive data isolation strategy supporting scenario-level, thread-level, and global-level data segmentation, enabling reliable parallel execution across 100+ concurrent threads with zero data contamination
- Built advanced reporting infrastructure using Allure with AspectJ listeners for real-time event capture, test step visualization, and intelligent failure diagnostics

AI & Intelligent Automation:

- Designed Agentic AI system leveraging Model Context Protocol (MCP) for intelligent test case generation with Playwright CLI command execution
- Implemented token reduction optimization using grep-based search strategies and deterministic code execution to reduce LLM API costs by 60%
- Architected snapshot storage system using CLI commands to capture application state, store externally, and retrieve on-demand, reducing context window bloat
- Engineered self-healing test framework with AI agents that detect UI changes and automatically update test logic based on application evolution

Test Execution & Scalability:

- Engineered enterprise UI/API automation frameworks using Java, Playwright, and REST Assured managing 5,000+ automated test scenarios
- Improved legacy Selenium framework execution success from 60% to 97% through Page Object Model refactoring, intelligent retry mechanisms, and parallel execution
- Developed comprehensive mobile automation suite with 70+ end-to-end Android and iOS scenarios using Appium with BDD Cucumber framework
- Integrated AppCircle REST APIs for CI/CD automation, automating build triggering, artifact retrieval, and deployment, reducing manual setup by 85%
- Collaborated with 40+ engineers across QA, Development, and DevOps to drive continuous automation innovation and framework optimization

TECHNICAL SKILLS

Languages: Java, SQL, Python, JavaScript

UI Automation: Playwright, Selenium, Appium, XPath, CSS Selectors, CLI Commands

API Testing: REST Assured, Postman, Query-based Data Fetching

Frameworks: BDD, Cucumber, TestNG, Maven, Page Object Model, Data-Driven Testing, Parallel Execution

AI & Advanced: Agentic AI, Model Context Protocol (MCP), Self-healing Tests, Intelligent Screen Scanning, Token Optimization

Reporting: Allure Report, AspectJ Listeners, SLF4J, Real-time Event Capture

DevOps & Tools: Jenkins, Git, Docker, JIRA, AppCircle API Integration, IntelliJ IDEA

Databases: SQL, MySQL, JDBC, Dynamic Query Execution

KEY ACHIEVEMENTS

- Designed intelligent screen-scanning framework reducing manual locator maintenance by 70% through dynamic label-ID mapping
- Execution success rate improved from 60% to 97% through architectural enhancements including data isolation and parallel execution strategies
- 5,000+ UI/API automated test scenarios with query-based data fetching across multiple environments
- 70+ end-to-end mobile automation scenarios (iOS/Android) with intelligent data isolation at scenario, thread, and global levels
- 60% reduction in LLM API costs through token optimization using grep-based search and deterministic code execution
- 40% reduction in test execution time through framework optimization and CLI-based deterministic operations
- Performance rating: 4/4 for technical contributions, architectural design, and innovation in automation frameworks

EDUCATION

Bachelor of Technology, Computer Science & Engineering | PES University, Bengaluru | 2020 - 2024

CERTIFICATIONS

- AWS Cloud Practitioner Essentials
- Version Control with Git (Coursera)